Removal of Specular Highlights in Endoscopic Images using 3D Gaussian Splatting

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Abstract. Specular highlights frequently appear in endoscopic images due to the structural characteristics of the co-located light source and camera, obscuring the color and texture of organ tissue and thereby hindering clinicians' observation and judgment, as well as degrading 3D scene reconstruction. Because specular highlights exhibit characteristics that vary with viewpoint position and intensity, they distort both tissue appearance and scene geometry when not explicitly handled. To reconstruct tissue appearance free of specular highlights from multi-view endoscopic images, a method is proposed that decomposes input images into a view-independent base structure and a view-dependent specular component within a 3D Gaussian Splatting-based scene representation. Specifically, specular highlights are estimated by combining an HSV-based prior with positive residuals that exceed the estimated base structure. A weighted reconstruction loss and decoupled optimization are applied to mitigate the influence of specular highlights on appearance and geometry reconstruction. Experimental results confirm that the proposed method reconstructs consistent tissue appearance in regions subject to strong specular highlights.

Keywords: specular highlight, 3D Gaussian Splatting, endoscopic reconstruction.

1 Introduction

Endoscopic images are indispensable for diagnosing and treating lesions within hollow organs such as the gastrointestinal tract, airways, and urogenital system. Their applications now include intraoperative navigation, lesion tracking, and the construction of patient-specific digital twins. Obtaining accurate 3D scene reconstruction from endoscopic images has thus become increasingly critical [1, 2]. However, stable 3D scene reconstruction remains challenging due to the limited field of view, non-rigid tissue deformation, instrument occlusion, and illumination variation. In particular, the co-located configuration of the light source and camera in endoscopes frequently produces strong specular highlights, which not only obscure the intrinsic color and texture of tissue but also distort appearance and geometric reconstruction, as their position and intensity vary with the viewpoint.

This work exploits the observation that the same tissue surface may be occluded by specular highlights from certain viewpoints while appearing relatively stable from others. Leveraging the complementarity of such multi-view observations, a 3DGS-based scene representation method is proposed that explicitly estimates and removes specular highlights from input images to reconstruct tissue appearance. Specifically, input images are decomposed into a view-independent base structure and a view-dependent specular component; the specular component is then removed from pixel values to guide tissue appearance learning, thereby reducing distortion in appearance and geometry reconstruction.

To this end, 3D Gaussian Splatting (3DGS) [3] is adopted as the scene representation. An HSV-based weak prior is introduced to prevent regions dominated by specular highlights from exerting excessive influence on base structure learning. Positive residuals unexplained by the base structure are further utilized to selectively estimate the specular component. The optimization paths of the two components are decoupled so that specular component learning does not directly affect base structure representation. Through this design, the stability of geometry and appearance reconstruction is enhanced, and tissue appearance with preserved anatomical structure is more reliably recovered.

2 Related Work

Specular Highlight Removal in Endoscopic Images. Learning-based work used GANs to remove specular highlights from single endoscopic images [4]. Adaptive RPCA was applied to image sequences to separate background and sparse specular components [5]. A two-stage framework combining highlight segmentation with partial convolution and attention-based inpainting was also proposed [6]. An unsupervised temporal learning approach for endoscopic video was further introduced, with analysis of its effect on downstream correspondence tasks [7]. Despite these advances, most prior methods operate on single images or 2D sequences and do not explicitly leverage multi-view information for joint specular removal and 3D reconstruction.

3DGS for Endoscopic Scene Reconstruction. 3DGS-based methods have been proposed for endoscopic scene reconstruction. Real-time dense reconstruction and tracking were achieved using Gaussian representation [8], and 3DGS was further extended to deformable tissue reconstruction [9, 10]. Gaussian Pancakes [11] improved geometric stability through depth and surface regularization, and PR-ENDO [12] incorporated physically based relightable rendering for view-dependent illumination. However, these methods mainly focus on geometry, deformation, or illumination modeling rather than explicitly separating specular components from input images. The proposed method distinguishes itself by explicitly disentangling the specular component within a 3DGS-based scene representation to reconstruct tissue appearance with suppressed specular highlight influence.

3 Method

3.1 Scene Representation

The proposed method takes as input a set of multi-view endoscopic images $\{I_{\mathrm{GT}}\}$. The base structure of the scene is modeled using a 3DGS-based representation, in which each Gaussian primitive is parameterized by position, scale, opacity, and appearance parameters, and is projected into images via differentiable rendering from a given viewpoint. Input images are regarded as the combined result of the base structure and a view-dependent specular component:

$$I_{GT} \approx I_{base} + I_{spec} \tag{1}$$

where I_{base} denotes the base structure of tissue with specular highlights removed, and I_{spec} denotes the specular component present in the input images. In this work, 3DGS serves as the scene representation for reconstructing I_{base}; the estimation of I_{spec} and the specular highlight removal procedure are described in the following sections.

3.2 Specular Estimation

I_{spec} is estimated from I_{GT} by jointly employing an HSV-based prior and brightness residuals relative to I_{base}. Since specular highlights are generally characterized by high brightness and low saturation, the prior is defined as:

$$P_{\mathrm{HSV}} = ((1 - S) \cdot V)^{\gamma} \tag{2}$$

where S and V denote the saturation and value channels, and γ is a hyperparameter controlling the sensitivity of the prior. Components in the input images exceeding I_{base} are then treated as the specular component:

$$R_{spec} = \mathrm{ReLU}\,(I^{GT} - I_{base}) \tag{3}$$

The loss for the specular component is accordingly defined as:

$$\mathcal{L}_{spec} = \| P_{HSV} \odot (R_{spec} - I_{spec}) \|_1 \tag{4}$$

This loss encourages I_{spec} to capture components in the input images that exceed I_{base}, with emphasis on regions exhibiting strong specular highlights. In addition, the optimization of the specular component is decoupled from that of I_{base} so that specular component learning does not directly influence base structure reconstruction.

3.3 Specular Highlight Removal

The estimated I_{spec} is utilized to remove specular highlights from input images, upon which I_{base} is reconstructed. However, since the estimation of specular highlight I_{spec} may be incomplete in regions with strong specular highlights, pixel-wise

weights are introduced to prevent such uncertainty from being directly reflected in I_{base} learning. These weights are defined using the HSV prior as follows:

$$w = \text{clamp}(1 - \alpha P_{\text{HSV}}, w_{\text{min}}, 1) \tag{5}$$

where α controls the influence of the prior, and w_{min} is a lower bound on training stability. This allows more reliable observations to take priority in I_{base} reconstruction. The reconstruction loss for I_{base} is defined as:

$$\mathcal{L}_{\text{base}} = \| w \odot \left((I_{\text{GT}} - I_{\text{spec}}) - I_{\text{base}} \right) \|_1 \tag{6}$$

This loss guides I_{base} to reconstruct tissue appearance using specular removed observations the reconstruction target. The total loss combines the two terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{base}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}} \tag{7}$$

λ_{spec} is a weighting factor that controls the relative contribution of the two terms. The final output I_{base} represents the tissue appearance of endoscopic images with specular highlights removed.

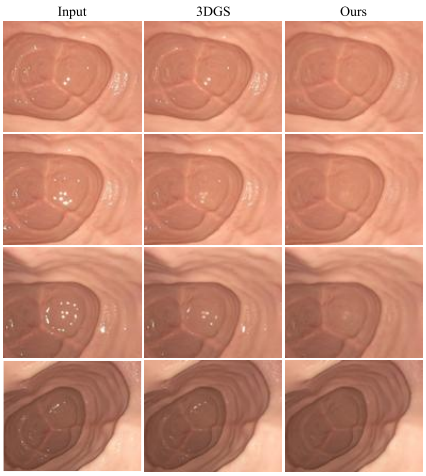


Fig. 1. Qualitative comparison of specular highlight removal results. Left: Original input, Center: 3DGS reconstruction, Right: Ours

4 Experiments and Results

Dataset. The Colonoscopy 3D Video Dataset (C3VD) [13] is employed for performance evaluation. C3VD is well-suited for analyzing endoscopic images with frequent specular highlights. This is because it physically simulates the colon's complex anatomical structure and the detailed specular properties of the mucosal surface. Preprocessing follows the EndoGSLAM [8] pipeline. The test set is constructed by extracting every 8th frame from each scene.

Metrics. To comprehensively validate reconstructed image quality, we adopt PSNR, SSIM [14], and LPIPS [15] as evaluation metrics. Quantitative evaluation is performed on regions excluding specular highlight areas to enable a more direct assessment of reconstructed tissue appearance.

Results. Fig. 1 presents a qualitative comparison between the proposed method and the baseline 3DGS. While the baseline reproduces brightness saturation and color distortion in strongly highlighted regions, the proposed method removes specular highlights and reconstructs consistent tissue color and texture.

Fig. 2 illustrates the specular highlight decomposition results of the proposed method. The separated specular highlights correspond to the specular highlight regions in the input images, and the overlay results confirm that the proposed method consistently disentangles them across views.

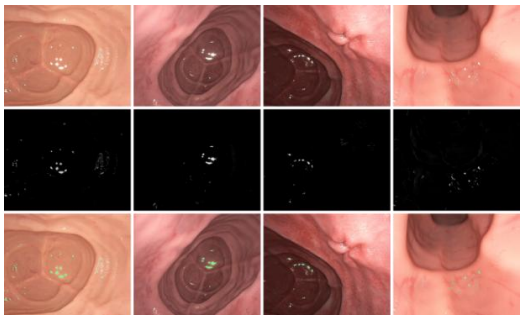


Fig. 2. Qualitative results of the proposed specular highlight separation on the C3VD dataset. first row: original input images, second row: separated specular highlights, third row: highlight regions (green) overlaid on original input images.

As shown in Table 1, the proposed method outperforms baseline 3DGS in PSNR and SSIM while maintaining comparable LPIPS scores. Explicitly decomposing and removing the specular component improves reconstruction accuracy without degrading tissue appearance. Notably, improvements in reconstruction metrics, even when excluding specular highlight regions, demonstrate that disentangling the specular component also helps appearance and geometry learning in surrounding regions.

Table 1. Quantitative results of reconstruction excluding specular highlight areas.

Method	PSNR	SSIM	LPIPS
3DGS	35.80	0.89	0.22
Ours	**35.88**	**0.90**	**0.22**

5 Conclusion

This paper proposes a method that explicitly estimates and removes specular highlights within a 3DGS-based multi-view scene representation to reconstruct tissue appearance. Input images are decomposed into a base structure and a specular com-

ponent, with an HSV-based prior and positive residuals used to selectively estimate the specular component. Decoupled optimization ensures that specular component learning does not degrade base structure reconstruction. Experimental results demonstrate that the proposed method reconstructs consistent tissue appearance even in strongly highlighted regions, with improved performance in both qualitative and quantitative evaluations, indicating the effectiveness of multi-view 3D representation over conventional 2D correction approaches.

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